Cálculo II

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O Teorema Fundamental do Cálculo – Parte 01

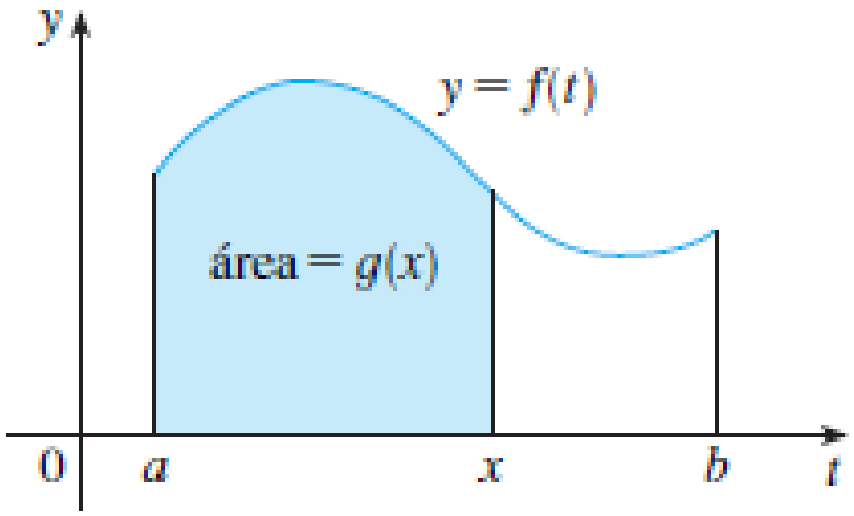
$$\frac{d}{dx}\left[\int_a^x f(t)\,dt\right] = f(x)$$

$= g(x)$

↑ os operadores de derivação e integração são "operadores"

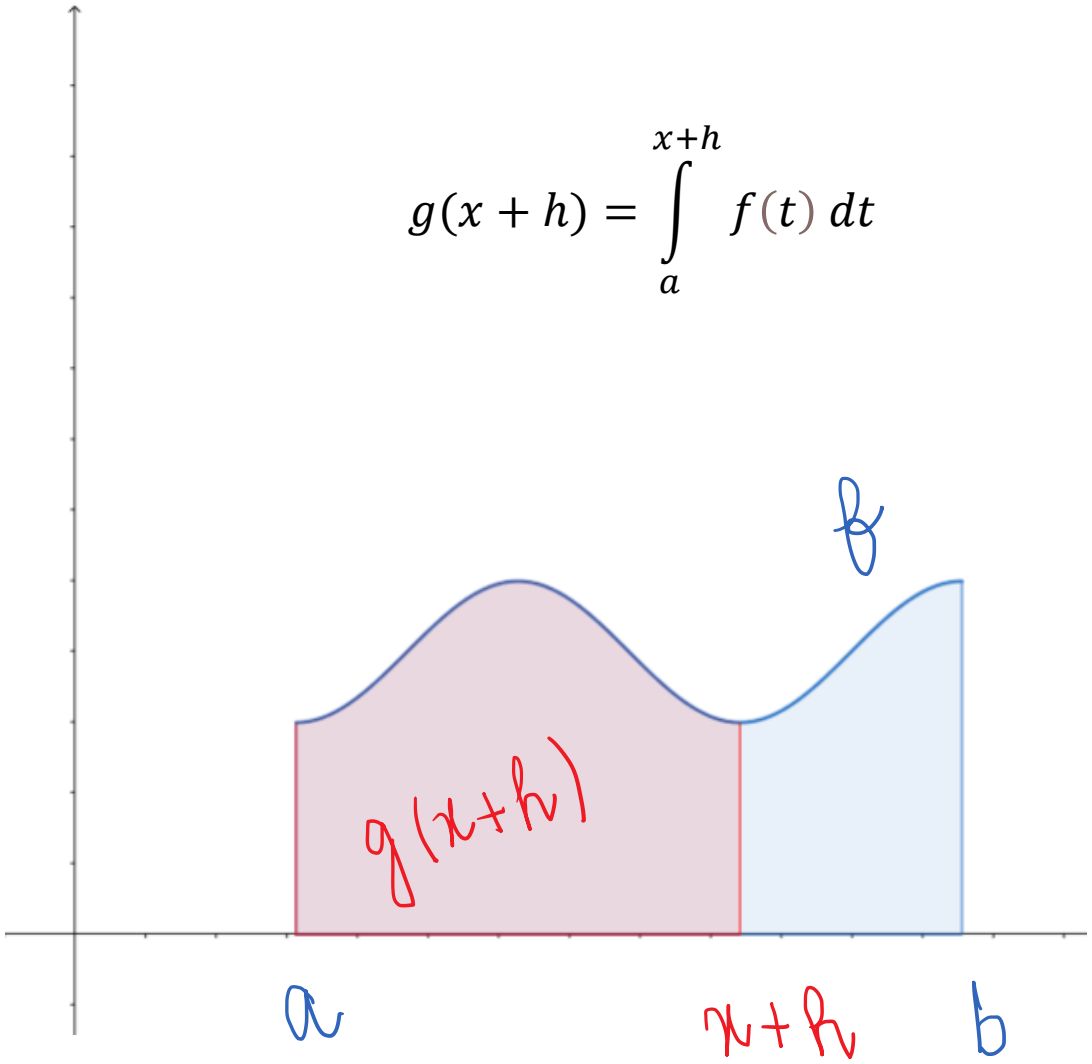
Revisando...

$$g(x) = \int_a^x f(t)\, dt$$



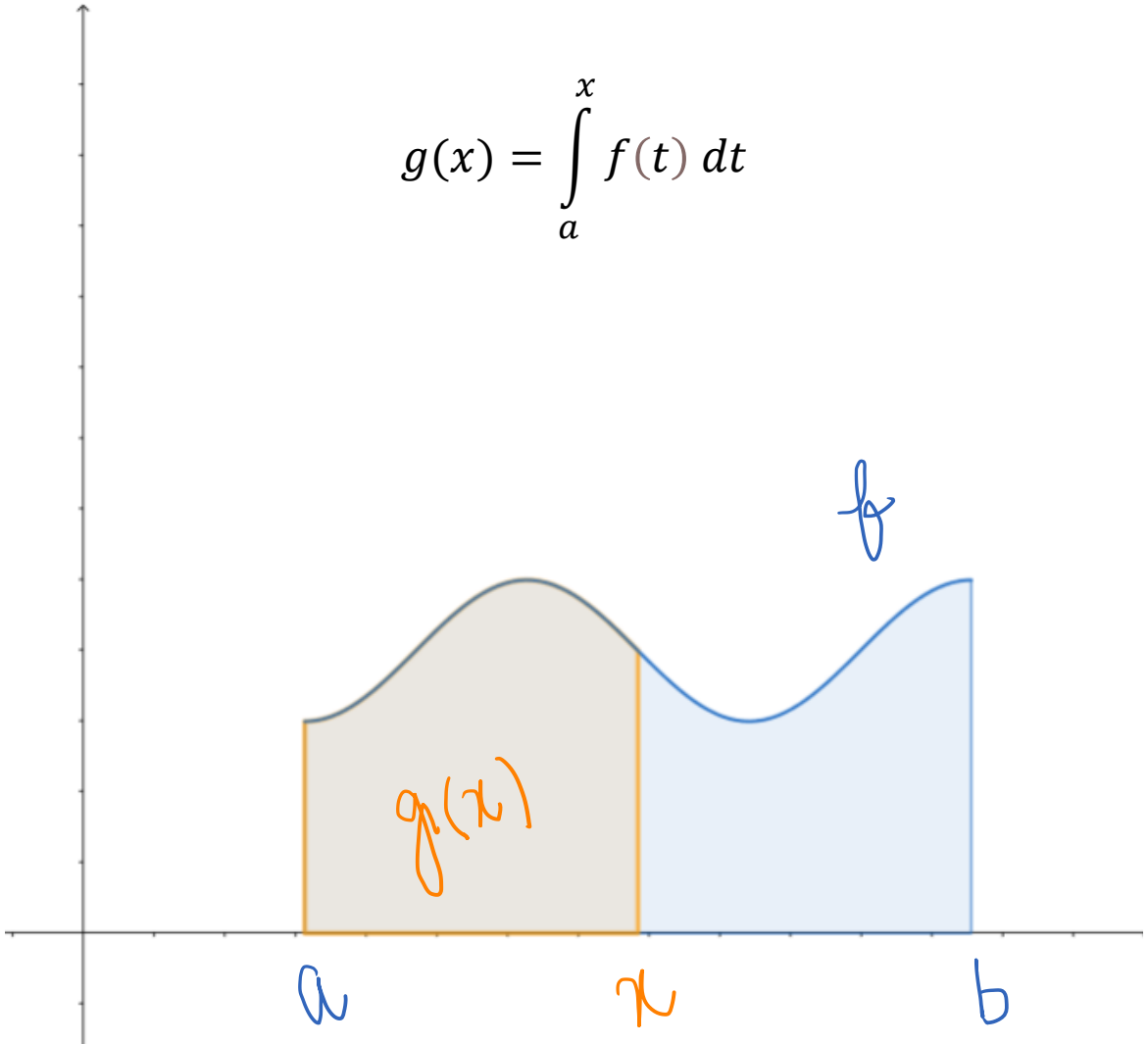
Entendendo...

$$g(x+h) = \int_{a}^{x+h} f(t)\, dt$$



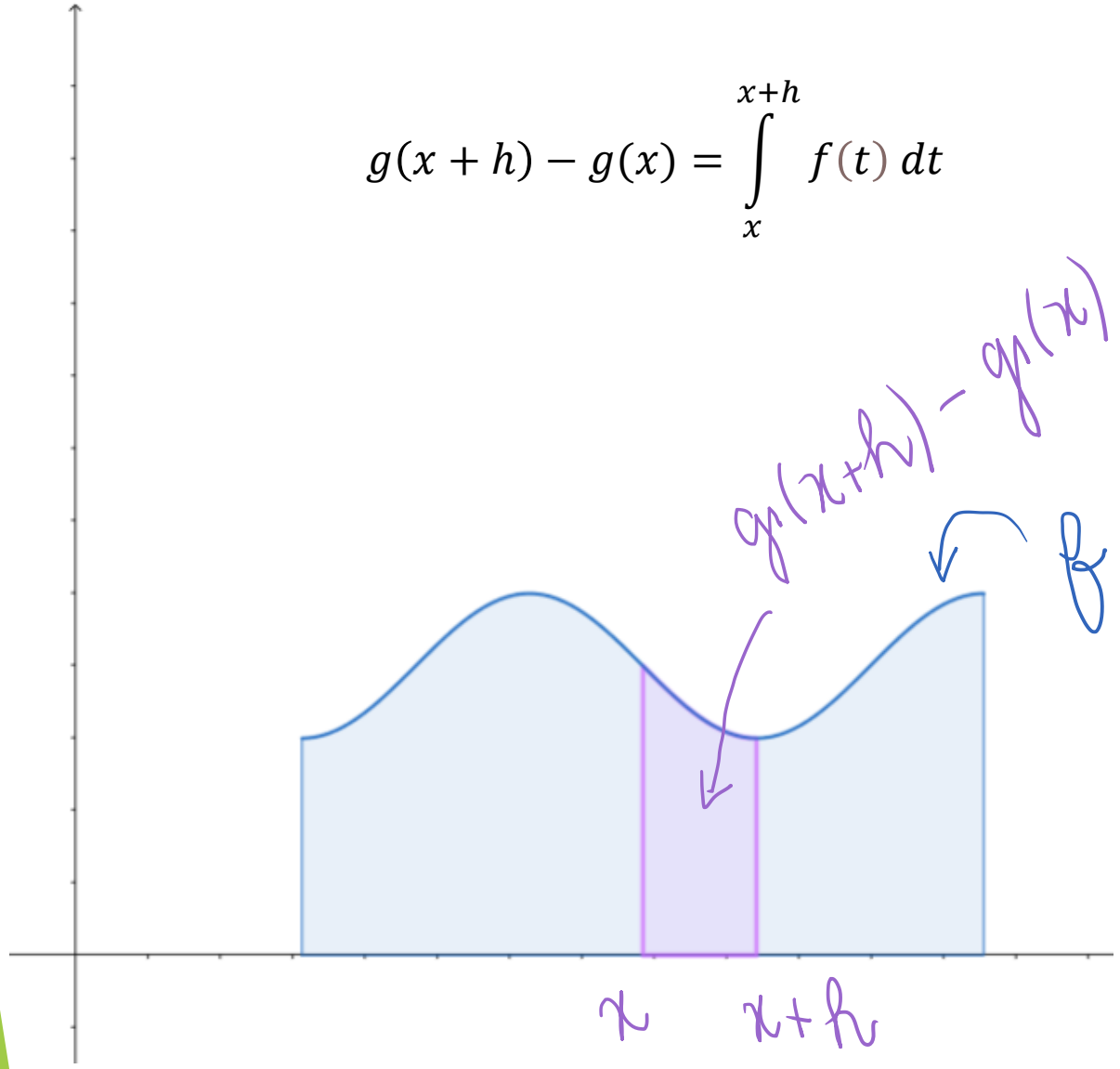
Entendendo...

$$g(x) = \int_a^x f(t)\,dt$$



Entendendo...

$$g(x+h) - g(x) = \int_{x}^{x+h} f(t)\, dt$$



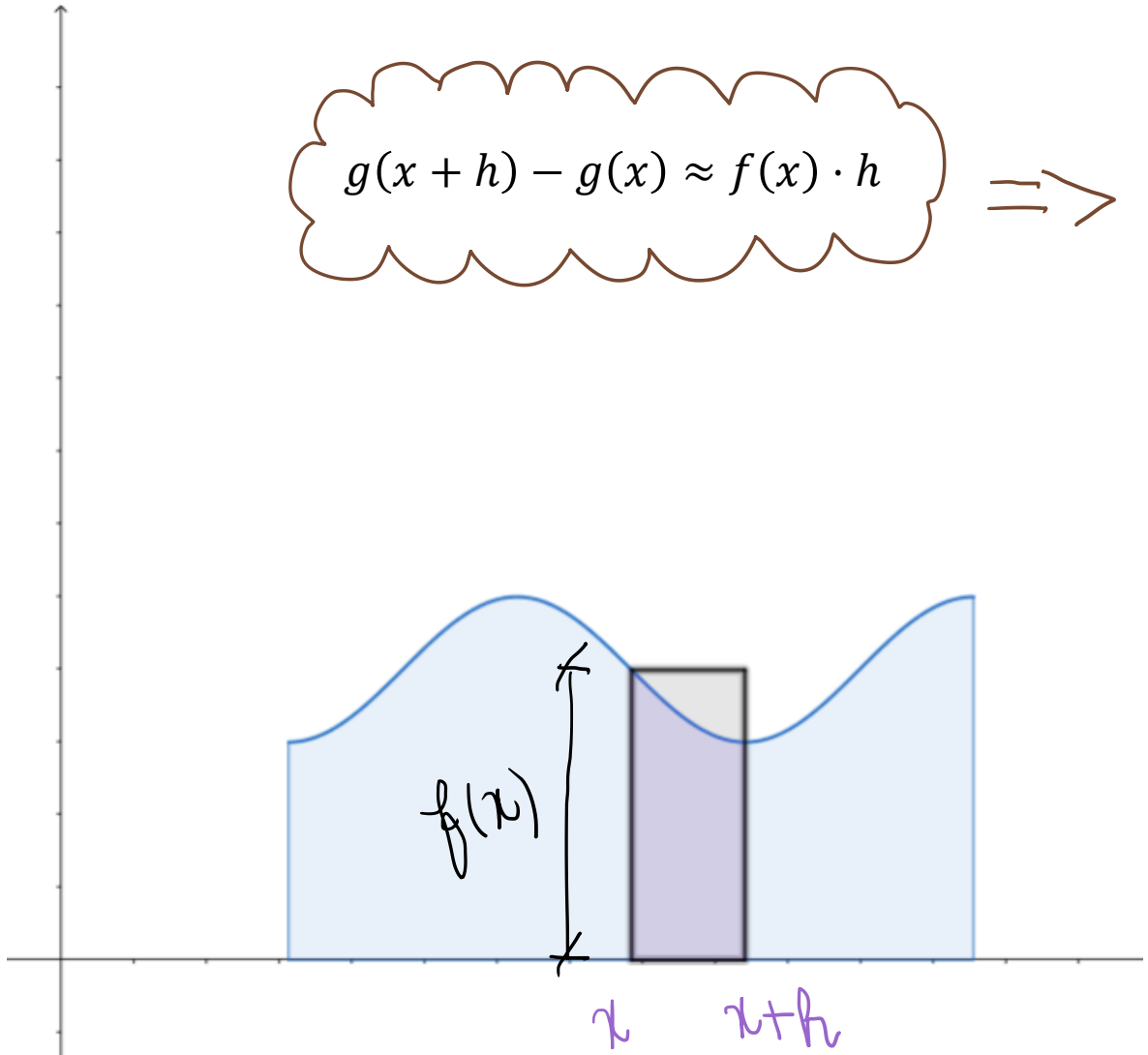
Entendendo...

$$g(x+h) - g(x) \approx f(x) \cdot h \Rightarrow \frac{g(x+h) - g(x)}{h} \approx f(x)$$

$$\downarrow$$

$g'(x)$,

quando $h \to 0$



Praticando...

$$g(x) = \int_1^x \frac{1}{t^3+1}\,dt$$

$= f(t)$

Temos que:

$$g'(x) = \frac{d}{dx}\left[\int_1^x \frac{1}{t^3+1}\,dt\right] = \frac{1}{x^3+1}$$

Praticando...

é problema!!!

$$g(x) = \int_1^{e^x} \ln t \, dt$$

$= f(t)$

Seja $u(x) = e^x$. Daí, $g(u(x)) = \int_1^u f(t)\,dt$.

Logo, $\frac{d}{dx} g(u(x)) = \frac{d}{du} g \cdot \frac{d}{dx} u = \frac{d}{du}\left[\int_1^u f(t)\,dt\right] \cdot \frac{d}{dx}(e^x) =$

regra da cadeia

ótimo!!!

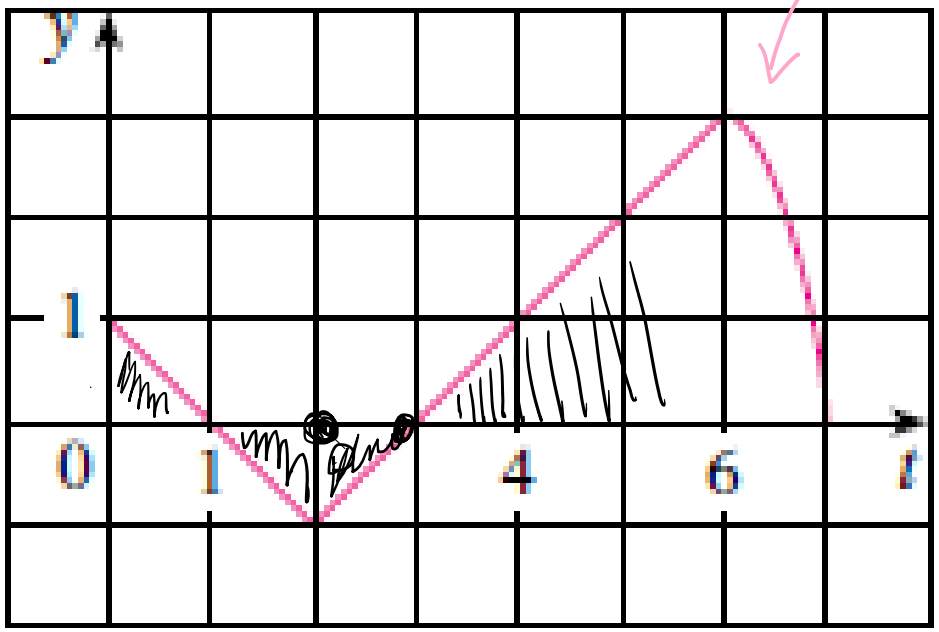
$= f(u) \cdot e^x = (\ln u)\, e^x = x e^x$

$= e^x$

Praticando...

$$g(x) = \int_0^x f(t)\ dt$$

gráfico de f



a) $g(1) = \int_0^1 f(t)\,dt = \frac{1}{2} \cdot 1 \cdot 1 = \frac{1}{2}$

b) $g(6) = \int_0^6 f(t)\,dt =$

não está abaixo do eixo t

$= \frac{1}{2} \cdot 1 \cdot 1 + \left(-\frac{1}{2} \cdot 2 \cdot 1\right) + \frac{1}{2} \cdot 3 \cdot 3 =$

$= \frac{1}{2} - 1 + \frac{9}{2} = 4$

Praticando...

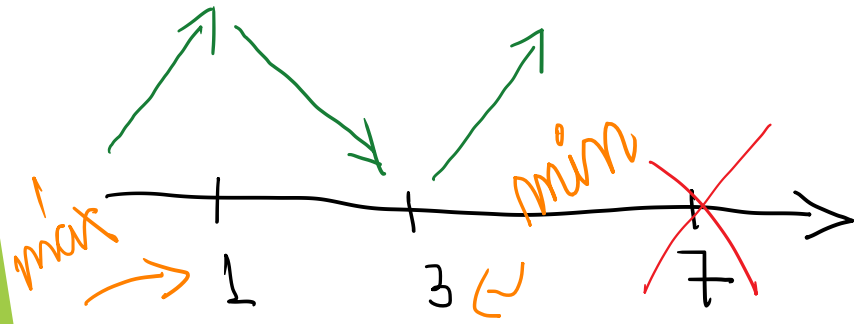
c) Onde g tem um máximo local?

$$g(x) = \int_0^x f(t)\ dt$$

Temos que: $\frac{d}{dx} g(x) = \frac{d}{dx}\left[\int_0^x f(t)\,dt\right] =$

$= f(x)$. Daí, $g'(x) = 0 \Rightarrow f(x) = 0$

$x=1,\ x=3,\ x=7$



sinal de $g' = f$

Se $x = c$ é um ponto crítico de g ($g'(c) = 0$ ou $\nexists\, g'(c)$) e g' troca de sinal em $x = c$, então $x = c$ é um ponto de extremo de g.

O Teorema Fundamental do Cálculo – Parte 02

via definição $= \lim_{n \to +\infty} \left[\sum_{i=1}^{n} f(x_i^*) \Delta x \right]$

$g(x) = \int_a^x f(t)\,dt$ é tal que $g'(x) = f(x)$, ou seja, g é uma primitiva de f.

$$\int_a^b f(x)\,dx = F(b) - F(a),$$

em que F é uma primitiva de f.

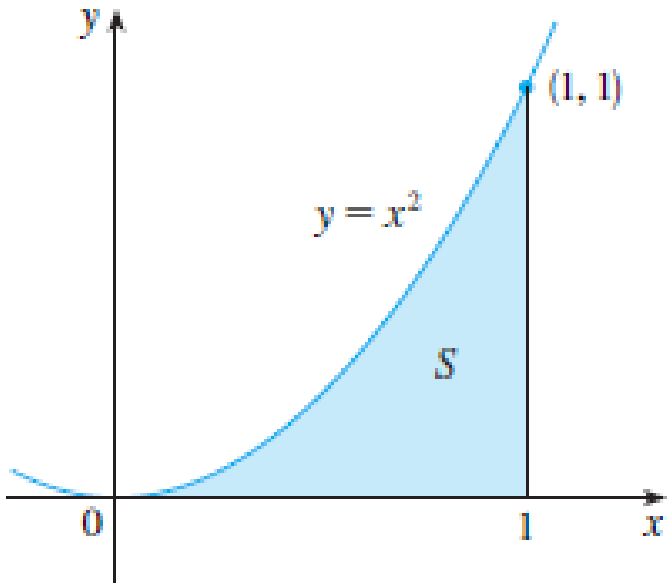
$F(x) = g(x) + C$ é a primitiva mais geral de f.

De fato, $F(b) - F(a) = [g(b) + C] - [g(a) + C] = g(b) - g(a) =$

$= \int_a^b f(t)\,dt - \cancel{\int_a^a f(t)\,dt} = 0 = \int_a^b f(t)\,dt$ $t = x$, $dt = dx$

Praticando...

$$\int_0^1 x^2\,dx = \frac{x^3}{3}\Big|_0^1 = \frac{1^3}{3} - \frac{0^3}{3} = \frac{1}{3}$$



Praticando...

$$\int_{-1}^{2} (x^3 - 2x)\,dx = \left(\frac{x^4}{4} - x^2\right)\Bigg|_{-1}^{2} = \left(\frac{2^4}{4} - 2^2\right) - \left[\frac{(-1)^4}{4} - (-1)^2\right] =$$

$$= 0 - \left(-\frac{3}{4}\right) = \frac{3}{4}.$$

Praticando...

$$\int_1^2 \frac{4+x^2}{x^3}\,dx = \left(4\,\frac{x^{-3+1}}{-3+1} + \ln|x|\right)\Bigg|_1^2 =$$

$$= \left(-2\cdot 2^{-2} + \ln|2|\right) - \left(-2\cdot 1^{-2} + \ln|1|\right) =$$

$$= -\frac{1}{2} + \ln 2 + 2 = \frac{3}{2} + \ln 2$$

$$\frac{4}{x^3} + \frac{x^2}{x^3} =$$

$$= 4x^{-3} + x^{-1}$$

Praticando...

$$\int_0^3 (2\sin x - e^x)\,dx = \left(-2\cos x - e^x\right)\Big|_0^3 =$$

$$= \left(-2\cos 3 - e^3\right) - \left(-2\underbrace{\cos 0}_{=1} - \underbrace{e^0}_{=1}\right) =$$

$$= -2\cos 3 - e^3 + 3$$

Praticando...

$$\int_1^2 (1+2x)^2\,dx = \left(x + 4\frac{x^2}{2} + 4\frac{x^3}{3}\right)\Bigg|_1^2 =$$

$$(1+2x)^2 = 1 + 4x + 4x^2$$

$$= \left(1 + 2\cdot 2^2 + \frac{4}{3}\cdot 2^3\right) - \left(1 + 2\cdot 1^2 + \frac{4}{3}\cdot 1^3\right) =$$

$= \ldots$ continue

Integrais Indefinidas

$$\int f(x)\,dx = F(x),$$

significa que $F'(x) = f(x)$.

Praticando...

$$\int \cos^2 \mathrm{x} \;\; dx = \frac{x}{2} + \frac{1}{4}\sin(2x) + C$$

Praticando...

$$\int \left(x^4 - \frac{1}{2}x^3 + \frac{1}{4}x - 2 \right) \, dx = ?$$

Praticando...

$$\int (\sec^2 x \ - 2e^x) \ dx = ?$$

Praticando...

$$\int_{1}^{4} \frac{\sqrt{x} - x}{x^4}\, dx = ?$$

Praticando...

$$\int_{1}^{2} (2x - 3) \cdot (4\mathrm{x}^2 + 1)\, dx = ?$$

Quais os significados das expressões?

$$\int_a^b \mathrm{v(t)}\, dt = ?$$

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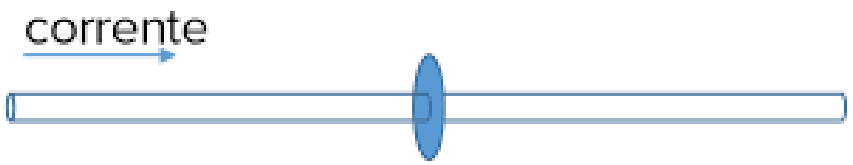

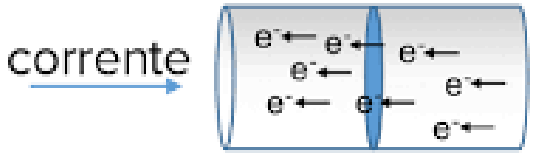

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$$\int_a^b \mathrm{r(t)}\, dt = ?$$

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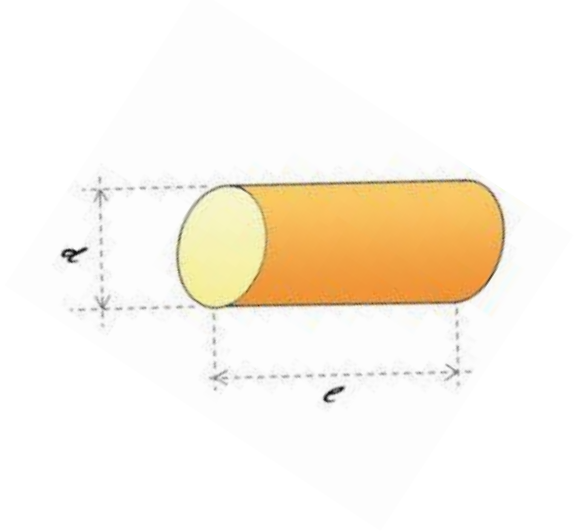
$$\int_a^b \rho(\mathrm{x})\, dx = ?$$



Generalizando...

$$\int_a^b f'(x)\,dx = f(b) - f(a)$$

A integral da taxa de variação de uma grandeza num intervalo

=

A variação total da grandeza neste intervalo